

FIELDGRAPH-NET: PROTOCOL-GUIDED NETWORK INTRUSION DETECTION WITH STRUCTURAL ANALYSIS

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ABSTRACT

Network intrusion detection must accommodate traffic from protocols absent during classifier training. We introduce FieldGraph-Net, a structural front end that uses predicted protocol identity to guide flow analysis. Message segmentation yields candidate fields used to construct a structural fingerprint with 24 components and a heterogeneous field graph. A graph classifier predicts protocol identity from the field graph, using protocol labels for supervision. The router sends flows assigned to the attack classifier's training protocol domain to that classifier and directs the remainder to structural clustering. A decision rule combines classifier alerts with structural risk indicators. On CIC IoT 2023, thresholds are selected from benign scores in the evaluation set to obtain realized false positive rates near 20%. Across five attack categories, FieldGraph-Net achieves a mean F1 of 0.5076, compared with 0.4323 for maximum softmax probability, the strongest baseline by mean F1. Baseline results are averaged over five seeds; FieldGraph-Net contributes one operating point per category. The framework extends analysis coverage beyond the attack classifier's protocol domain and achieves the highest mean F1 among the evaluated methods.

Index Terms— Network intrusion detection, protocol identification, heterogeneous graph neural networks, structural fingerprinting, structural clustering

1. INTRODUCTION

Graph neural networks (GNNs) are widely used for network intrusion detection [1, 2, 3]. Their deployment often assumes that the protocol distribution observed during training remains representative [4]. Campus and industrial networks challenge this assumption as services and protocol mixtures change over time. A classifier can still assign a confident attack label to a flow from an unfamiliar protocol. Identifying whether that protocol belongs to the classifier's training domain is therefore a distinct step in assessing its prediction.

Existing approaches address different parts of this problem. Retraining requires examples from the new setting, while learning from positive and unlabeled data requires an observed positive class [5]. Open set classifiers provide a rejection mechanism for unknown attacks [6]. Out-of-distribution (OOD) detectors use maximum softmax probability (MSP) [7], calibrated confidence [8], or Mahalanobis distance [9] to assess compatibility with training data. Distribution shifts can impair these scores in network intrusion detection [10], as discussed more broadly in anomaly detection [11]. Local Outlier Factor [12] and Isolation Forest [13] estimate anomalies without attack labels. These scores quantify deviations from

reference data; protocol identification provides complementary information about the traffic being assessed.

Protocol reverse engineering extracts candidate fields from message structure [14], and hierarchical format inference refines this representation [15]. Protocol clustering also groups unknown traffic through similarities in inferred specifications [16]. These studies motivate using field organization to guide intrusion detection. Structural extraction can operate without attack labels, while protocol classification can use protocol labels to learn recurring formats. Linking the resulting identity signal to a routing decision allows unfamiliar traffic to receive a separate structural assessment.

We propose FieldGraph-Net to implement this separation. A structural fingerprint and a heterogeneous field graph encode each segmented message. Field Protol GNN predicts one of twelve protocol labels, and a domain router uses protocol identity to select the subsequent analysis path. Flows within the attack classifier's training protocol domain receive attack predictions. Flows outside that domain enter structural clustering, which identifies unusual message organization and associated risk indicators. A decision rule then assigns an explicit state to each flow.

Our contributions are summarized as follows:

- We combine structural fingerprints with heterogeneous field graphs to obtain protocol identity using protocol supervision, without requiring attack labels for this stage.
- We introduce routing by protocol domain to combine attack classification with structural assessment of flows outside the classifier's training domain.
- We evaluate the system on CIC IoT 2023 and CICIDS2017. The comparison across five attack categories yields a mean F1 of 0.5076, and the ablation characterizes the contributions of classifier scores, structural scores and the full rule.

2. PROPOSED METHOD

Figure 1 presents three stages: structural representation, protocol identification, and routing with decision fusion. Protocol predictions assign flows to the classifier's training protocol domain or to structural analysis; the latter are termed OOD flows here.

2.1. Structural Representation

BCDG Field Segmentation. We use Gaussian filtering of bit congruence differences (BCDG), the structural signal introduced by NEMESYS [14]. Bit congruence is defined as

$$C(i) = \frac{1}{8} \sum_{j=0}^7 \mathbf{1}[b_i[j] = b_{i+1}[j]], \quad 1 \leq i < n, \quad (1)$$

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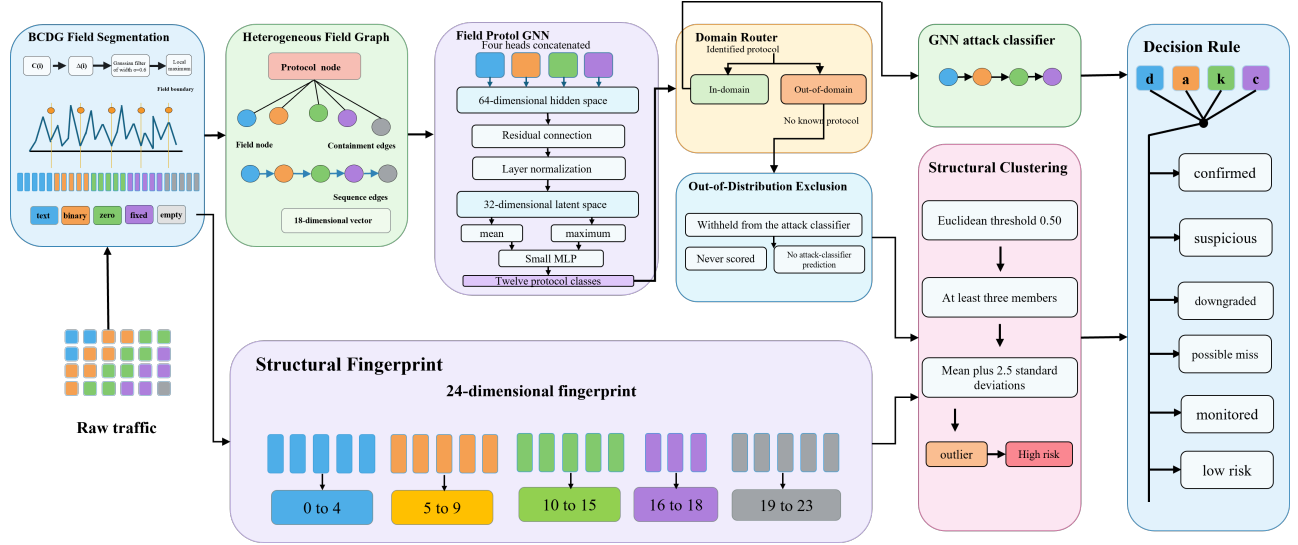


Fig. 1. Overview of FieldGraph-Net. Message segmentation supplies structural fingerprints and heterogeneous field graphs. Field Protol GNN predicts protocol labels for domain routing. Flows within the classifier’s training protocol domain receive attack predictions; other flows enter structural clustering. The six labels shown in the decision interface reduce to four reachable states under the routing constraint.

where $C(i)$ is the fraction of equal bits in adjacent bytes and $1[\cdot]$ is the indicator function. The message contains n bytes; $b_i[j]$ denotes bit j of byte i , with bit indices from zero to seven. The indicator equals one for matching bits and zero otherwise, so the expression averages eight comparisons. The first difference is $\Delta(i) = C(i+1) - C(i)$ for $1 \leq i \leq n-2$, where $\Delta(i)$ measures the change between consecutive congruences and i , C and n retain these definitions. Gaussian smoothing uses $\sigma = 0.6$, where σ is the kernel standard deviation in byte positions. Our local segmentation rule selects maxima of the smoothed differences when the preceding segment contains at least two bytes. The last segment extends to the payload boundary, and adjacent printable ASCII segments are merged. Fields retain their offsets, lengths and implementation labels: text, binary, zero, fixed or empty.

Structural Fingerprint. Each message is encoded as $\phi = (\phi_0, \dots, \phi_{23})$, where ϕ is the structural fingerprint and the subscripted entries are its 24 feature components. Components 0 to 4 describe the field count and the mean, standard deviation, minimum and maximum field lengths. Components 5 to 9 contain the proportions of the five field labels. Components 10 to 15 encode Shannon entropy, the distinct byte ratio, and the first four payload bytes. Components 16 to 18 encode the transport protocol and source and destination ports. Each port is transformed as $q(p) = \log_2(p+1)/16$, where p is an integer port number from 0 to 65535, $q(p)$ is its normalized value, and $\log_2$ is the base two logarithm. Components 19 to 23 form a histogram of field lengths: 1, 2, 3 to 4, 5 to 8, and at least 9 bytes.

2.2. Field Graph Protocol Identification

Heterogeneous Field Graph. Each graph contains one message summary node, termed the protocol node, and one node per field. The field vector has 18 components describing offset, length, log length, initial bytes, relative position, boundary indicators, entropy, byte diversity and field type. The protocol node also has 18 components, summarizing message length, field statistics, field types, entropy, byte diversity, ports and transport protocol. Containment

edges connect the protocol node to each field, and sequence edges connect adjacent fields. Reversing both relations produces four directed relation types.

Field Protol GNN. Protocol classification uses two heterogeneous graph attention layers with each four heads [17]. The update with concatenated heads is

$$\mathbf{h}_v^{(\ell+1)} = \parallel_{\nu=1}^4 \rho \left(\sum_{r \in \mathcal{R}_v} \sum_{u \in \mathcal{N}_r(v)} \alpha_{vu}^{(\ell, r, \nu)} \mathbf{W}_r^{(\ell, \nu)} \mathbf{h}_u^{(\ell)} \right), \quad (2)$$

where v is the receiving node, u is a source node, and ℓ indexes layers. The indices r and ν denote a directed relation and an attention head, respectively. The vectors $\mathbf{h}_u^{(\ell)}$ and $\mathbf{h}_v^{(\ell+1)}$ are the input and output node embeddings, respectively. The matrix $\mathbf{W}_r^{(\ell, \nu)}$ projects source embeddings for the specified relation, layer and head. The set $\mathcal{R}_v$ contains the relations entering v , and $\mathcal{N}_r(v)$ contains the corresponding source nodes. The weight $\alpha_{vu}^{(\ell, r, \nu)}$ controls their contributions to the sums; ρ is an elementwise activation, and $\parallel$ concatenates the four heads. The layers produce embeddings of 64 and 32 components, respectively. A residual connection and layer normalization follow the first layer. Concatenating the protocol node embedding with projected mean and maximum pools of field embeddings forms the message representation. A multilayer perceptron maps this representation to twelve protocol classes using protocol labels for supervision.

2.3. Domain Routing and Decision Fusion

Domain Router. The router compares the identified protocol with those represented in the attack classifier’s training domain. A port lookup provides a fallback when a protocol identity is unavailable. Domain membership and protocol recognition are distinct: a recognized protocol can still lie outside the attack classifier’s training domain.

Out-of-Distribution Exclusion. The router sends OOD flows to structural analysis. The attack classifier supplies scores only for

flows routed to its domain. This separation defines the coverage of the two analysis paths.

Structural Clustering. Greedy clustering assigns each OOD fingerprint to the nearest existing centroid within Euclidean distance 0.50, creating a cluster if none qualifies. For clusters with at least three members, the outlier rule is

$$\delta_x = \|\phi_x - \mu_C\|_2, \quad \delta_x > m_C + 2.5s_C, \quad (3)$$

where x indexes a message in cluster C and ϕ_x denotes its structural fingerprint. The centroid μ_C defines the distance δ_x , and $\|\cdot\|_2$ denotes the Euclidean norm. The quantities m_C and s_C are the mean and standard deviation of the cluster reference distances used at scoring time. A cluster receives a high risk flag if it contains an outlier or combines a risk indicator, such as high entropy, with loose or moderate cohesion.

Decision Rule. Under domain routing, the active decision rule has four mutually exclusive outcomes:

$$\Phi(d, a, k, c) = \begin{cases} \text{confirmed,} & d \wedge a, \\ \text{possible miss,} & \neg d \wedge \neg k \wedge c, \\ \text{monitored,} & \neg d \wedge \neg k \wedge \neg c, \\ \text{low risk,} & \text{otherwise,} \end{cases} \quad (4)$$

where Φ is the output state, and $\wedge$ and $\neg$ denote logical conjunction and negation. The Boolean indicators d , a , k and c denote training domain membership, a classifier attack prediction, recognized protocol identity and a high risk cluster, respectively. Routing imposes $a \leq d$, where a uses zero as a bookkeeping value for OOD flows and d retains its domain indicator meaning. Consequently, the interface labels suspicious and downgraded are inactive because both require an attack prediction on an OOD flow. The confirmed state denotes a classifier alert, and possible miss denotes a structural alert for an unrecognized protocol.

3. EXPERIMENTS

3.1. Experimental Settings

Datasets and Baselines. The CIC IoT 2023 evaluation [18] uses 70 packet capture (PCAP) chunks containing 3233 flows. This subset includes benign traffic and six of the base detector’s seven attack output classes. Its denial of service (DoS) class has no samples. CICIDS2017 [19] provides a separate evaluation corpus excluded from training. The local extractor computes the 82 flow features specified by the GNN4ID feature pipeline [1] directly from PCAP files. The base detector was retrained with its reference pipeline using NFStream [20]. Feature values and flow counts differ between the two extraction pipelines, introducing a preprocessing difference between training and evaluation. All compared methods use the same local extractor and base detector. The OOD baselines are MSP [7], ODIN [8] with temperature calibrated on benign flows from the training domain, and Mahalanobis scoring [9] with class means and shared covariance. These guards use the existing classifier and its training information without additional attack annotations for OOD calibration.

Metrics and Implementation. Thresholds are selected from benign scores to minimize the absolute difference between the realized false positive rate (FPR) and 20%, with ties resolved by the smaller threshold. We report recall, F1 and FPR; the score AUC is the area under the receiver operating characteristic curve. Table 2 averages baseline results across five seeds and reports one operating point per

attack category for FieldGraph-Net. Evaluation uses Python 3.10.11, PyTorch 2.13.0, PyTorch Geometric 2.8.0, scikit-learn 1.7.2, NumPy 2.2.6 and SciPy 1.15.3 on an Intel Core i9-13980HX running Windows 11. Detector retraining uses an NVIDIA RTX 4060 with PyTorch 2.13.0+cu126; evaluation runs on the CPU.

3.2. Results and Analysis

Simulation Experiment. Table 1 compares alert rates in documented CICIDS2017 attack windows with the reference rate on the same day. All extracted flows from the selected captures were retained. Enrichment is the ratio of the window alert rate to the corresponding reference rate, providing a temporal measure of alert concentration. The evaluation uses the published attack schedule to define windows; reliable alignment between the released flow labels and the locally extracted flows was not established.

Table 1. CICIDS2017 evaluation using all extracted flows. Alert rates in documented attack windows are compared with the reference alert rate on the same day. Enrichment is the ratio of these rates.

Day	Window	Alert	Enrich.
Wed	DoS Slowloris	0.5008	1.81×
	DoS Slowhttptest	0.4104	1.48×
	DoS Hulk	0.6882	2.48×
	DoS GoldenEye	0.5166	1.86×
	Heartbleed	0.2588	0.93×
Fri	Botnet ARES	0.2038	0.74×
	PortScan, rules on	0.2833	1.03×
	PortScan, rules off	0.8120	2.96×
	DDoS LOIT	0.0959	0.35×
Reference, Wed / Fri		0.2772 / 0.2739	1.00×

Five windows showed enrichment from 1.48 to 2.96 times the reference rate: Slowloris, Slowhttptest, Hulk, GoldenEye and PortScan with firewall rules disabled. PortScan with rules enabled yielded 1.03 times the reference rate, a small numerical increase. Heartbleed, Botnet ARES and DDoS LOIT yielded ratios of 0.93, 0.74 and 0.35, respectively. Thus, alert concentration varied across attack windows and firewall conditions.

Structural clustering responds to deviations in message organization. Repeated messages with similar structure can remain close to a cluster centroid even when their aggregate activity is malicious. This property motivates complementary features for traffic volume and temporal behavior. The window comparisons identify settings for examining these complementary signals.

Comparative Experiment. Table 2 reports F1 for five attack categories. FieldGraph-Net achieved the highest reported mean F1, 0.5076, compared with 0.4323 for MSP, 0.3710 for the raw detector and 0.3287 for ODIN. The difference from MSP was 0.0753. These means summarize performance across the listed attack categories at the reported operating points.

Table 2. F1 across five attack categories at a realized FPR near 20%. Thresholds use benign scores. Baseline entries average five seeds; Ours uses one operating point per category.

Attack	GNN-raw	MSP	ODIN	Mahal.	Ours
DNS.Spoof	0.2747	0.3560	0.1667	0.5390	0.4792
Mirai	0.4060	0.4506	0.2873	0.3580	0.6230
SqlInjection	0.3519	0.3574	0.3688	0.4136	0.4226
DictBruteForce	0.4425	0.3883	0.2444	0.4465	0.5676
Recon-PortScan	0.3801	0.6093	0.5764	0.3831	0.4457
Mean	0.3710	0.4323	0.3287	0.4280	0.5076

Relative to the strongest baseline in each category, FieldGraph-Net improved F1 by 0.1724 on Mirai and 0.1211 on dictionary brute force. Its F1 on SQL injection was 0.4226, exceeding the strongest baseline by 0.0090. On DNS spoofing, it trailed Mahalanobis scoring by 0.0598; on reconnaissance port scanning, it trailed MSP by 0.1636. The reported point estimates therefore favor FieldGraph-Net in three categories and a baseline in two.

Threshold selection uses benign observations from the evaluation set to match the reported FPR target. The comparison therefore characterizes performance at that operating point. An independently calibrated deployment threshold can produce a different realized FPR. The baseline entries average five seeds, while each FieldGraph-Net entry represents one operating point for the corresponding category.

Ablation Experiment. Table 3 reports the classifier probability (A1), structural cluster score (A3), full decision rule (A4), and a fixed routing rule (A2). A1, A3 and A4 operate at FPRs of 0.200, 0.196 and 0.206, respectively. A2 has a single operating point, with an FPR of 0.868.

Table 3. Reported ablation results for a corpus of 3233 flows. A1, A3 and A4 target an FPR near 20%. A2 reports its fixed operating point. AUC is not reported for rule outputs.

Variant	Score	FPR	Recall	F1	AUC
A1	classifier prob.	0.200	0.2759	0.4204	0.6063
A3	cluster score	0.196	0.2378	0.3735	0.4973
A4	full rule	0.206	0.3696	0.5252	n/a
A2	fixed rule	0.868	0.8968	0.8726	n/a

A4 recorded recall of 0.3696 and F1 of 0.5252. The corresponding A1 entries were 0.2759 and 0.4204, giving numerical differences of 0.0937 in recall and 0.1048 in F1 at an FPR difference of 0.006. A3 recorded recall of 0.2378, F1 of 0.3735 and AUC of 0.4973. Its score discrimination was close to the chance reference of 0.5, while the full rule had the highest recall and F1 among the rows targeting 20% FPR. A2 attained recall of 0.8968 together with a substantially higher FPR of 0.868.

The evaluation pool contains 3233 flows. A1 supplies classifier scores for 984 flows, while routing sends the remaining 2249 flows to structural analysis. The full rule combines classifier and structural information at the reported operating point.

4. CONCLUSION

FieldGraph-Net combines protocol identification, domain routing and structural clustering for network intrusion detection. Protocol supervision provides an identity signal without attack labels for the structural front end. Routing then connects attack classification with a separate assessment of unfamiliar traffic. Across five CIC IoT 2023 attack categories, the system achieved mean F1 of 0.5076, compared with 0.4323 for MSP. The full rule recorded recall of 0.3696 and F1 of 0.5252 at an FPR of 0.206. CICIDS2017 evaluation showed that alert concentration varies across attack windows, motivating complementary analysis of traffic volume and timing. The results characterize how routing by protocol combines classifier predictions with structural evidence at explicit operating points.

5. ACKNOWLEDGMENT

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6. ETHICAL STANDARDS STATEMENT

Our evaluation uses existing packet captures from CIC IoT 2023 and CICIDS2017 for research on network intrusion detection. Reuse should follow dataset terms and limit access to payloads and network identifiers that may expose sensitive information. Protocol inference can aid both defense and reconnaissance, so use of FieldGraph-Net should remain within explicitly authorized networks. The observed false alarms warrant analyst review before traffic is blocked or devices are isolated.

7. OPEN SCIENCE

To support reproducibility, we provide an anonymized artifact package containing the source code, experimental scripts, model checkpoints, and selected aggregate results. The raw CIC IoT 2023 and CICIDS2017 captures are not included and must be obtained from their respective sources. The package is available at the anonymized artifact repository.

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